

A MIMO Model for an Axon Under Chemical–Mechanical Coupling - a Note

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Abstract

Signal propagation in excitable biological media is traditionally modeled as a purely electrochemical phenomenon. In neuronal axons, this view is embodied in the Hodgkin–Huxley framework and its spatial extension, the cable equation. However, a growing body of experimental and theoretical work suggests that electrical activity in axons is intrinsically coupled to mechanical deformation through osmotic pressure, cytoskeletal elasticity, and membrane tension. In this work, we derive from first principles a coupled chemical–mechanical model for an axon, formulated as a distributed multiple–input multiple–output (MIMO) system. Starting from Ohm’s law, Kirchhoff’s laws, and basic mechanical balance principles, we derive a cable equation on a curved one–dimensional manifold and couple it rigorously to an axial mechanical field. No prior knowledge of differential geometry is assumed. The resulting framework unifies electrical conduction, chemical excitability, and mechanical deformation in a single mathematical structure.

1 Introduction

The classical theory of nerve signal propagation rests on the Hodgkin–Huxley equations and their spatial generalization, the cable equation [1]. These models successfully explain a wide range of electrophysiological observations, yet they idealize the axon as a rigid, straight electrical cable embedded in an inert medium. In reality, axons are soft, deformable structures, immersed in fluid, whose membranes swell and contract during electrical activity. Mechanical effects accompanying action potentials have been observed experimentally for decades, including transient changes in axonal diameter and surface tension [2].

Recent hydromechanical models in plant and neural systems further demonstrate that mechanical and chemical signals can propagate together and influence one another [3]. These observations motivate the development of a unified theoretical framework in which electrical, chemical, and mechanical processes are treated on equal footing.

In this paper, we construct such a framework from first principles. We emphasize derivation over phenomenology and guide the reader step by step from elementary physical laws to a coupled system of partial differential equations. The final model has the structure of a distributed MIMO system, with electrical and mechanical fields acting as both inputs and outputs.

2 Electrical Conduction in an Axon: From Ohm to Cable

2.1 Ohm's Law in a Cylindrical Conductor

We begin with the most basic electrical principle: Ohm's law. In differential form, Ohm's law states that the current density $\mathbf{J}$ is proportional to the electric field $\mathbf{E}$,

$$\mathbf{J} = \sigma \mathbf{E}, \quad (1)$$

where σ is the electrical conductivity of the medium.

In an axon, current flows predominantly along the longitudinal direction. We idealize the axon as a slender cylindrical conductor of radius a and length much greater than a . Let x denote the coordinate along the axis. The electric field is then approximately $E_x = -\partial V / \partial x$, where $V(x, t)$ is the transmembrane voltage.

Integrating the current density over the cross-sectional area $A = \pi a^2$, the axial current becomes

$$I(x, t) = -\sigma A \frac{\partial V}{\partial x}. \quad (2)$$

2.2 Kirchhoff's Current Law

Kirchhoff's current law states that the net current entering a small segment must equal the current leaving it plus any current lost through the membrane. Consider a small segment of axon between x and $x + \Delta x$. The difference in axial current is

$$I(x, t) - I(x + \Delta x, t) \approx -\frac{\partial I}{\partial x} \Delta x. \quad (3)$$

This difference must equal the membrane current per unit length, denoted $i_m(x, t)$, times Δx :

$$-\frac{\partial I}{\partial x} = i_m. \quad (4)$$

Substituting the expression for I , we obtain

$$\frac{\partial}{\partial x} \left(\sigma A \frac{\partial V}{\partial x} \right) = i_m. \quad (5)$$

2.3 Capacitance and Ionic Currents

The membrane current has two components: a capacitive current and an ionic (conductive) current. The capacitive current per unit length is

$$i_c = C_m \frac{\partial V}{\partial t}, \quad (6)$$

where C_m is the membrane capacitance per unit length.

The ionic current represents the flow of ions through channels and is written generically as

$$i_{\text{ion}} = g(V, \mathbf{w})(V - V_{\text{eq}}), \quad (7)$$

where g is a voltage- and state-dependent conductance and $\mathbf{w}$ denotes gating variables.

Thus,

$$i_m = C_m \frac{\partial V}{\partial t} + i_{\text{ion}}. \quad (8)$$

Combining all terms yields the classical cable equation,

$$C_m \frac{\partial V}{\partial t} = \frac{\partial}{\partial x} \left(\sigma A \frac{\partial V}{\partial x} \right) - i_{\text{ion}}. \quad (9)$$

3 Geometry: From Straight Cables to Curved Axons

3.1 Why Geometry Matters

Real axons are not perfectly straight. They bend, twist, and deform under mechanical stress. To account for this, we replace the straight coordinate x with a more general coordinate that follows the axon's centerline.

3.2 Curves and Coordinates

Let the axon be represented by a smooth curve $\gamma(s)$ embedded in three-dimensional space, where s denotes arc length. By definition,

$$\left| \frac{d\gamma}{ds} \right| = 1. \quad (10)$$

The electrical potential is now a function $V(s, t)$ defined along the curve. The spatial derivative must be taken along the curve, which naturally leads to the concept of a *metric*.

3.3 Metric and Line Element

Along a one-dimensional curve, the infinitesimal distance is

$$ds^2 = g(s) d\xi^2, \quad (11)$$

where ξ is an arbitrary coordinate and $g(s)$ is the metric coefficient. For arc length, $g = 1$, but mechanical deformation can modify this relation.

The key consequence is that spatial derivatives must be weighted by the metric. The correct generalization of the second derivative is the Laplace–Beltrami operator,

$$\Delta V = \frac{1}{\sqrt{g}} \frac{\partial}{\partial \xi} \left(\sqrt{g} g^{-1} \frac{\partial V}{\partial \xi} \right). \quad (12)$$

3.4 Cable Equation on a Manifold

Replacing the ordinary second derivative in the cable equation by the Laplace–Beltrami operator, we obtain

$$C_m \frac{\partial V}{\partial t} = \frac{1}{\sqrt{g}} \frac{\partial}{\partial \xi} \left(\sqrt{g} D \frac{\partial V}{\partial \xi} \right) - i_{\text{ion}}, \quad (13)$$

where $D = \sigma A$ is the axial conductance.

This equation describes electrical propagation along a curved, deformable axon.

4 Mechanical Dynamics of the Axon

4.1 Axon as a Deformable Medium

We now introduce mechanics. Let $u(s, t)$ denote the axial displacement of the axon. The axial strain is

$$\varepsilon = \frac{\partial u}{\partial s}. \quad (14)$$

4.2 Balance of Linear Momentum

Newton's second law for a one-dimensional elastic medium gives

$$\rho \frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial s} \left(E \frac{\partial u}{\partial s} \right) - \gamma \frac{\partial u}{\partial t} + f_{\text{chem}}, \quad (15)$$

where ρ is the linear mass density, E is an effective elastic modulus, γ is a damping coefficient, and f_{chem} is a force density generated by chemical activity.

4.3 Electrochemical Forcing

Electrical activity changes ion concentrations, generating osmotic pressure and membrane tension. To leading order, we assume the chemical forcing depends on voltage,

$$f_{\text{chem}} = \beta G(V), \quad (16)$$

where β is a coupling constant.

5 Coupling Electrical and Mechanical Fields

Mechanical deformation modifies the metric g , which in turn modifies electrical diffusion. To first order,

$$g \approx 1 + 2 \frac{\partial u}{\partial s}. \quad (17)$$

Thus, electrical conduction depends on u , while mechanics is driven by V .

The coupled system becomes

$$C_m \frac{\partial V}{\partial t} = \frac{1}{\sqrt{g(u)}} \frac{\partial}{\partial s} \left(\sqrt{g(u)} D \frac{\partial V}{\partial s} \right) - i_{\text{ion}}(V, \mathbf{w}) + \alpha \frac{\partial u}{\partial s}, \quad (18)$$

$$\rho \frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial s} \left(E \frac{\partial u}{\partial s} \right) - \gamma \frac{\partial u}{\partial t} + \beta G(V). \quad (19)$$

6 MIMO Interpretation

The state vector

$$\mathbf{y} = \begin{pmatrix} V \\ u \end{pmatrix} \quad (20)$$

evolves under mutual coupling. Voltage acts as an input to the mechanical system, while deformation acts as an input to the electrical system. This constitutes a distributed MIMO system with two inputs and two outputs.

7 Discussion and Outlook

The framework derived here unifies electrical, chemical, and mechanical signaling in axons. It generalizes the classical cable equation to curved, deformable geometries and embeds it naturally within continuum mechanics. The model provides a foundation for studying waveform sharpening, threshold modulation, and bidirectional signal–mechanics feedback.

Future work will explore traveling wave solutions, stability analysis, and numerical simulations, as well as reductions connecting this framework to poroelastic and hydromechanical models in other biological systems [3].

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